

CS5275 – The Algorithm Designer's Toolkit
(S2 AY2025/26)

Lecture 15:

Boolean functions – Goldreich–Levin algorithm

Goldreich–Levin theorem

Goldreich–Levin Theorem:

Suppose we are given:

- a parameter $0 < \tau \leq 1$
- query access to a function $f : \{-1,1\}^n \rightarrow \{-1,1\}$

There is an algorithm that computes a list L of subsets of $[n]$ such that with probability $1 - \delta$:

- **Completeness:** $|\hat{f}(U)| \geq \tau \rightarrow U \in L$
- **Soundness:** $U \in L \rightarrow |\hat{f}(U)| \geq \frac{\tau}{2}$

The running time of the algorithm is $\text{poly}\left(n, \frac{1}{\tau}\right) \cdot O\left(\log \frac{1}{\delta}\right)$.

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Spectral concentration $\rightarrow$ efficient learning

(even without the knowledge of the specific M sets)

Corollary: Suppose the Fourier spectrum of $f : \{-1,1\}^n \rightarrow \{-1,1\}$ is $(\epsilon/4)$ -concentrated on at most M sets. Then f can be learned in $\text{poly}\left(n, \frac{1}{\epsilon}, M\right) \cdot O\left(\log \frac{1}{\delta}\right)$ time with accuracy ϵ and error probability δ .

Proof of the corollary

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$$\sum_{S \notin L} \hat{f}(S)^2 \leq \underbrace{M \cdot \tau^2}_{\text{at most } M \text{ sets}} + \underbrace{\frac{\epsilon}{4}}_{\text{the remaining sets}} \leq \frac{\epsilon}{2}$$

$\tau = \sqrt{\frac{\epsilon}{4M}}$

Proof of the corollary

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$$\tau = \sqrt{\frac{\epsilon}{4M}}$$

$$\sum_{S \notin L} \hat{f}(S)^2 \leq M \cdot \tau^2 + \frac{\epsilon}{4} \leq \frac{\epsilon}{2}$$

Fourier spectrum of f is $\epsilon/2$ -concentrated on L .

f can be learned in $\text{poly}\left(n, \frac{1}{\epsilon}, |L|\right) \cdot O\left(\log \frac{1}{\delta}\right)$ time.

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Proof of the Goldreich–Levin theorem

Goldreich–Levin Theorem:

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- **Completeness:** $|\hat{f}(U)| \geq \tau \rightarrow U \in L$
- **Soundness:** $U \in L \rightarrow |\hat{f}(U)| \geq \frac{\tau}{2}$

The running time of the algorithm is $\text{poly}\left(n, \frac{1}{\tau}\right) \cdot O\left(\log \frac{1}{\delta}\right)$.

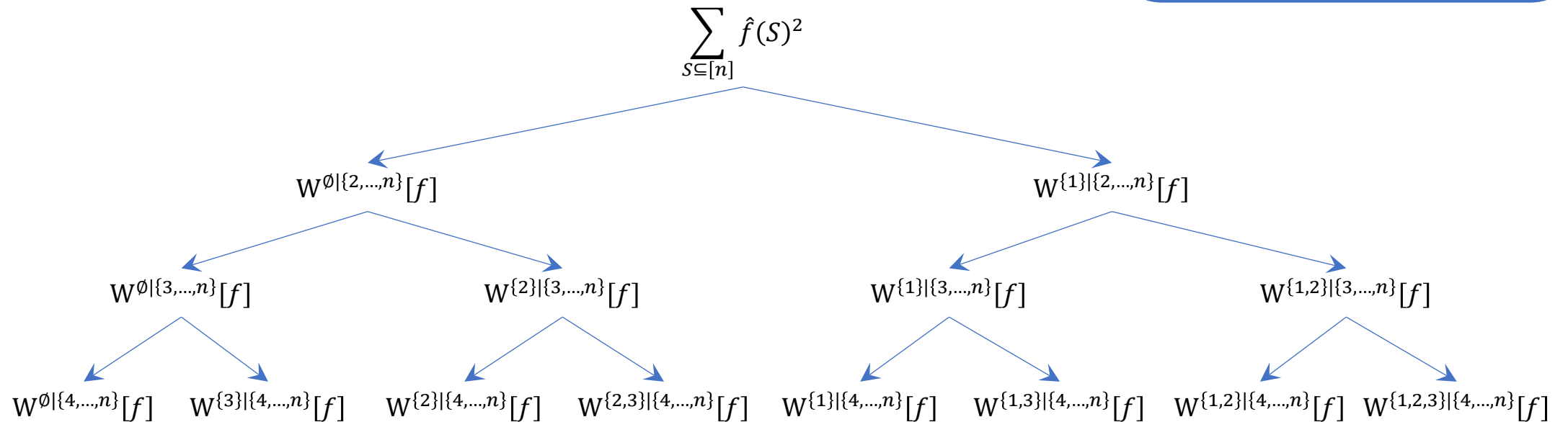


Next: Prove this theorem.

Restricted Fourier weights

- $f : \{-1, 1\}^n \rightarrow \mathbb{R}$
- $S \subseteq J \subseteq [n]$
- $\bar{J} = [n] \setminus J$
- We write:

$$W^{S|\bar{J}}[f] = \sum_{T \subseteq \bar{J}} \hat{f}(S \cup T)^2$$

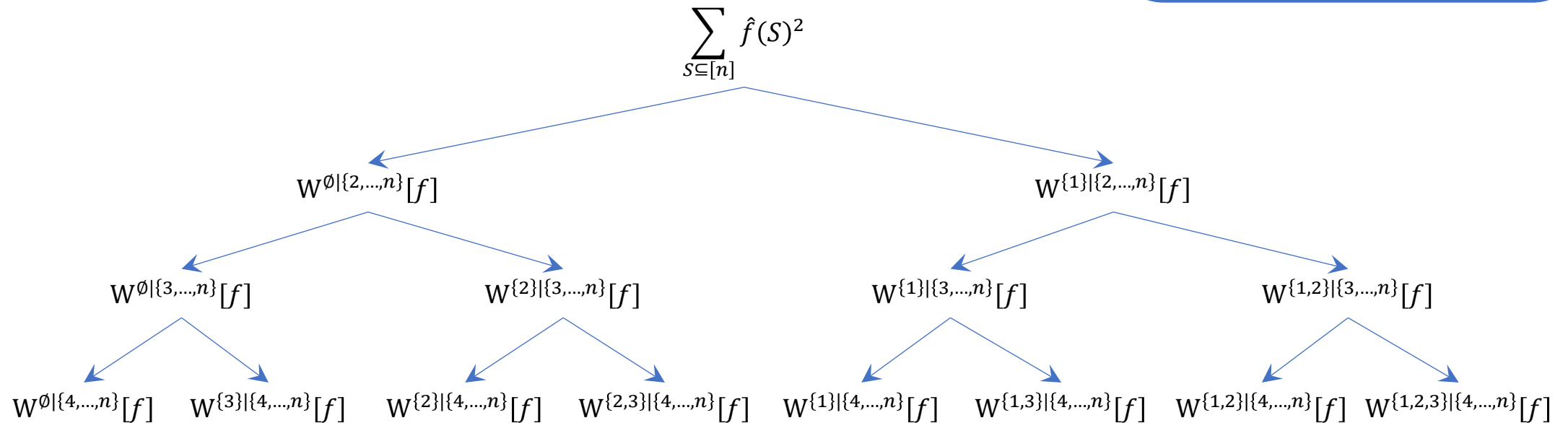


Estimation algorithm

Lemma: $W^{S|\bar{J}}[f]$ can be estimated using $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ queries of f with additive error $\pm \epsilon$ with probability $1 - \delta$.

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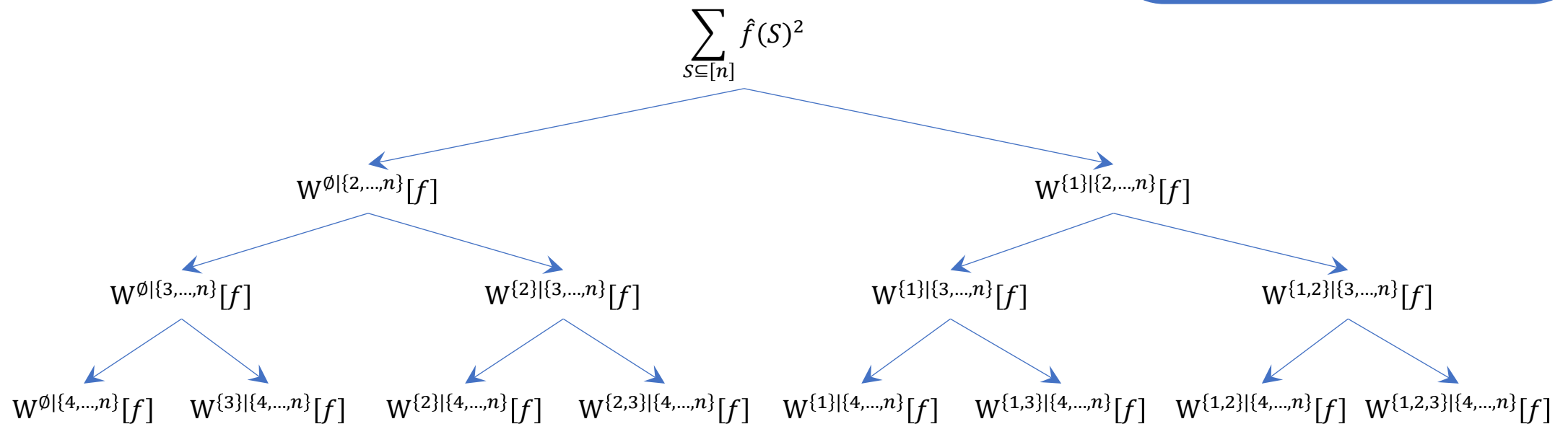
Estimation algorithm

Lemma: $W^{S|\bar{J}}[f]$ can be estimated using $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ queries of f with additive error $\pm \epsilon$ with probability $1 - \delta$.

Next: Using it to prove the Goldreich–Levin Theorem.

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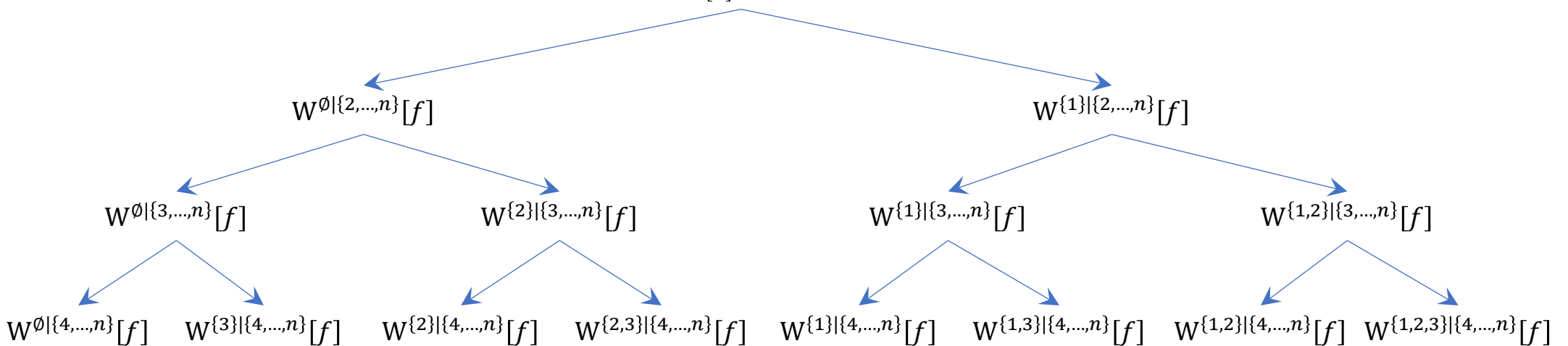
Goldreich–Levin Algorithm

Algorithm:

- For each node $W^{S|\bar{J}}[f]$:
 - Estimate its value with additive error $\pm \frac{\tau^2}{4}$.
 - If $\left(\text{estimated value} \geq \frac{\tau^2}{2}\right)$, search the two children.
- $L = \left\{ S : W^{S|\emptyset}[f] \text{ is a searched leaf node with estimated value} \geq \frac{\tau^2}{2} \right\}$.

$$\hat{f}(S)^2$$

Start here: $\sum_{S \subseteq [n]} \hat{f}(S)^2 = 1$



Lemma: $W^{S|\bar{J}}[f]$ can be estimated using $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ queries of f with additive error $\pm \epsilon$ with probability $1 - \delta$.

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Correctness

Algorithm:

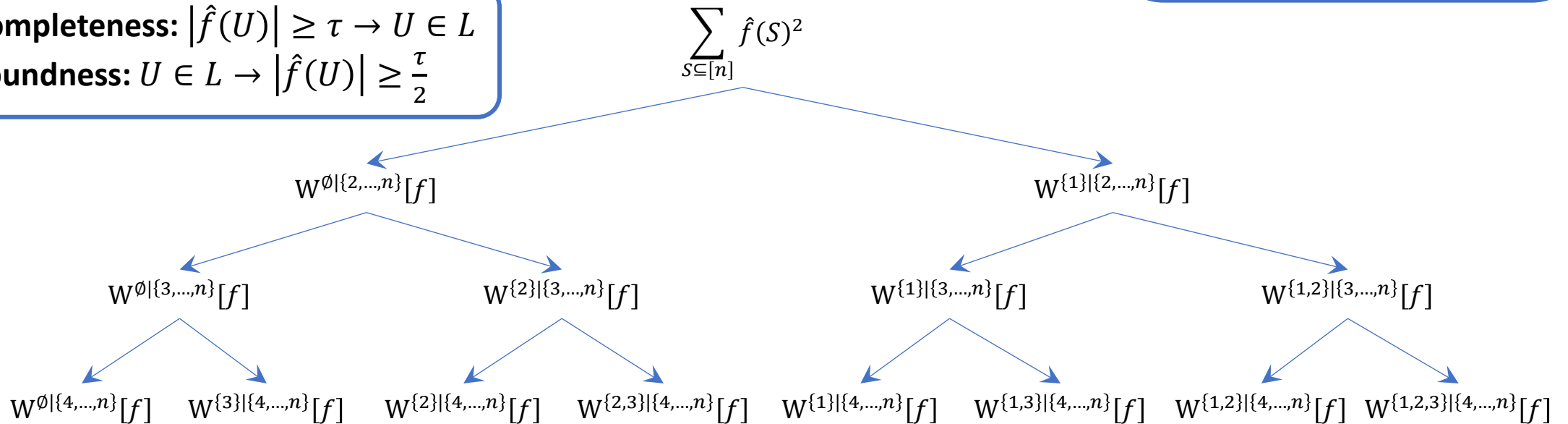
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How many times is this lemma applied?

Time complexity

Algorithm:

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If this is applied for t times, then the error probability for each application needs to be $\frac{\delta}{t}$.

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$T' =$ the subtree induced by the nodes $W^{S|\bar{J}}[f] \geq \frac{\tau^2}{4}$

The set of nodes searched is within the nodes in T' and their children.

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Number of leaves $\leq \frac{4}{\tau^2}$

Number of nodes $\leq n \cdot \frac{4}{\tau^2}$

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How many times is this lemma applied?

$O\left(\frac{n}{\tau^2}\right)$ times

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It remains to prove this lemma:

Proof of the lemma

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Input: $x = (y, z)$

$$f(y, z) = \sum_{S \subseteq J, T \subseteq \bar{J}} \hat{f}(S \cup T) \chi_{S \cup T}(y, z) = \sum_{S \subseteq J, T \subseteq \bar{J}} \hat{f}(S \cup T) \chi_S(y) \chi_T(z)$$

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For any given
 $S \subseteq J$:

Claim: $g(z) = \mathbb{E}_{y \sim \{-1, 1\}^J} [f(y, z) \chi_S(y)] = \sum_{T \subseteq \bar{J}} \hat{f}(S \cup T) \chi_T(z)$

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Next: Using the claim
to prove the lemma.

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$$\begin{aligned} W^{S|\bar{J}}[f] &= \sum_{T \subseteq \bar{J}} \hat{f}(S \cup T)^2 \\ &= \mathbb{E}_{z \sim \{-1, 1\}^{\bar{J}}} [g(z)^2] \\ &= \mathbb{E}_{z \sim \{-1, 1\}^{\bar{J}}} \left[\mathbb{E}_{y \sim \{-1, 1\}^J} [f(y, z) \chi_S(y)] \cdot \mathbb{E}_{y' \sim \{-1, 1\}^J} [f(y', z) \chi_S(y')] \right] \end{aligned}$$

Parseval

Lemma: $W^{S|\bar{J}}[f]$ can be estimated using $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ queries of f with additive error $\pm \epsilon$ with probability $1 - \delta$.

- $f : \{-1, 1\}^n \rightarrow \mathbb{R}$
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Lemma: $W^{S|\bar{J}}[f]$ can be estimated using $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ queries of f with additive error $\pm \epsilon$ with probability $1 - \delta$.



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Repeat for $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ times and take average:

- Sample y, y', z and calculate $f(y, z) \chi_S(y) f(y', z) \chi_S(y')$

Proof of the claim

Notation: Coordinate partition: $S \subseteq J$ $T \subseteq \bar{J}$
Input: $x = (\mathbf{y}, \mathbf{z})$

$$f(\mathbf{y}, \mathbf{z}) = \sum_{S \subseteq J, T \subseteq \bar{J}} \hat{f}(S \cup T) \chi_{S \cup T}(\mathbf{y}, \mathbf{z}) = \sum_{S \subseteq J, T \subseteq \bar{J}} \hat{f}(S \cup T) \chi_S(\mathbf{y}) \chi_T(\mathbf{z})$$

Claim: $g(\mathbf{z}) = \mathbb{E}_{\mathbf{y} \sim \{-1, 1\}^J} [f(\mathbf{y}, \mathbf{z}) \chi_S(\mathbf{y})] = \sum_{T \subseteq \bar{J}} \hat{f}(S \cup T) \chi_T(\mathbf{z})$

$$\begin{aligned} g(\mathbf{z}) &= \mathbb{E}_{\mathbf{y} \sim \{-1, 1\}^J} [f(\mathbf{y}, \mathbf{z}) \chi_S(\mathbf{y})] \\ &= \mathbb{E}_{\mathbf{y} \sim \{-1, 1\}^J} \left[\sum_{S' \subseteq J, T \subseteq \bar{J}} \hat{f}(S' \cup T) \chi_{S'}(\mathbf{y}) \chi_T(\mathbf{z}) \cdot \chi_S(\mathbf{y}) \right] \end{aligned}$$

Proof of the claim

Notation: Coordinate partition: $S \subseteq J$ $T \subseteq \bar{J}$
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Application

- The original motivation of Goldreich–Levin theorem:
 - Constructing a **pseudorandom generator** from a **one-way permutation**.

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A function $g : \mathbb{F}_2^k \rightarrow \mathbb{F}_2^m$ for some $m > k$ whose output on a random input looks unpredictable to any efficient algorithm.

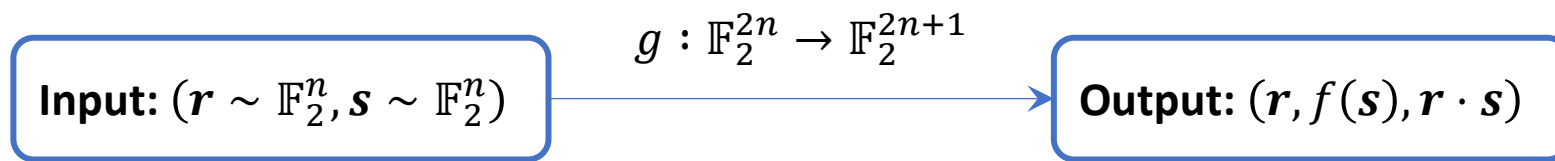


A bijective function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ that is easy to compute on all inputs but hard to invert for almost all input.



The construction:

Application



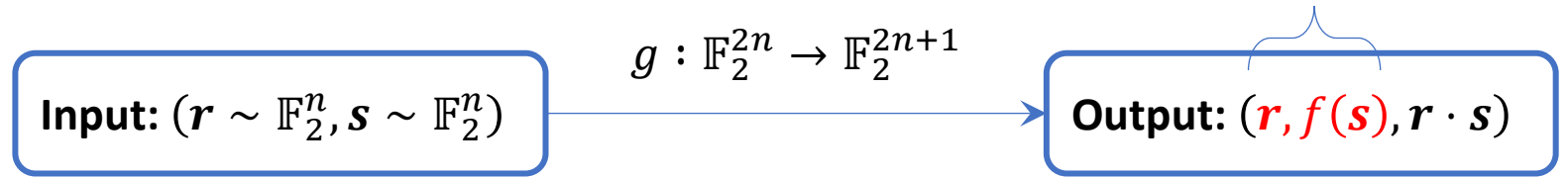
- The original motivation of Goldreich–Levin theorem:
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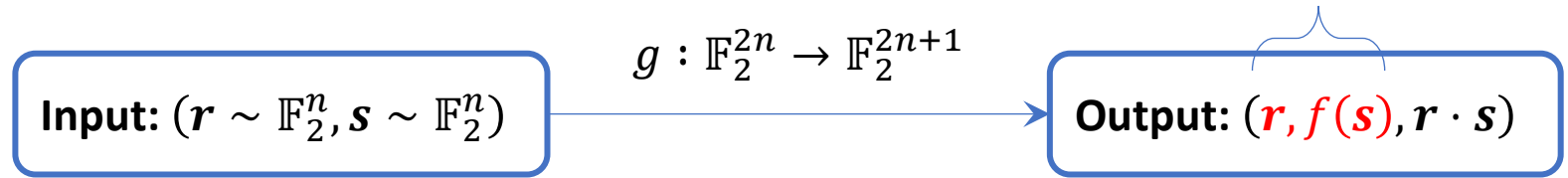
Assuming $\mathbf{r} \sim \mathbb{F}_2^n, \mathbf{s} \sim \mathbb{F}_2^n$, this part is already uniformly random.

Proof sketch



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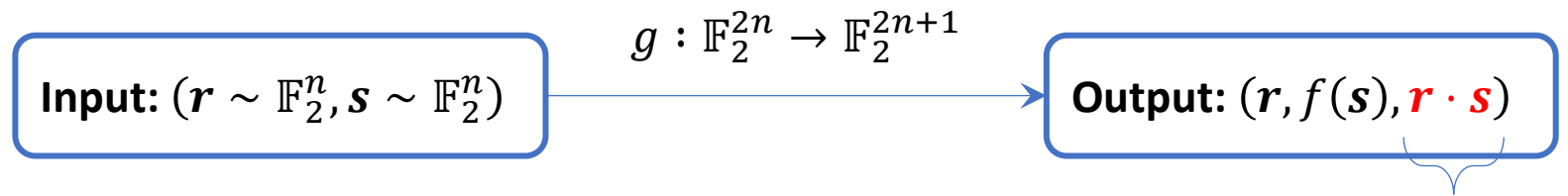


In fact:

$(\mathbf{r}, \mathbf{s})$ is pseudorandom $\rightarrow (\mathbf{r}, f(\mathbf{s}))$ is pseudorandom

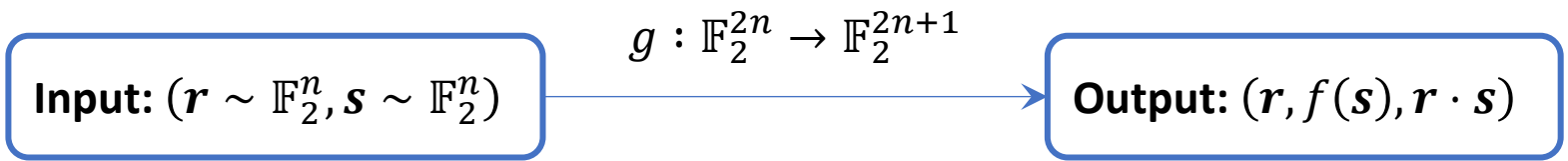
If there is an algorithm that can distinguish $(\mathbf{r}, f(\mathbf{s}))$ from a uniformly random $2n$ -bit string, then the same algorithm can also be used to distinguish $(\mathbf{r}, \mathbf{s})$ from a uniformly random $2n$ -bit string, as we assume that $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is easy to compute on all inputs.

Proof sketch



Next: Show that the final bit is also hard to predict.

Proof sketch



Suppose an efficient algorithm A can predict bit $\mathbf{r} \cdot \mathbf{s}$ from $\mathbf{r}, f(\mathbf{s})$ with non-negligible advantage:

$$\Pr_{\mathbf{r}, \mathbf{s}}[A(\mathbf{r}, f(\mathbf{s})) = \mathbf{r} \cdot \mathbf{s}] \geq \frac{1}{2} + \gamma$$

Proof sketch

Input: $(\mathbf{r} \sim \mathbb{F}_2^n, \mathbf{s} \sim \mathbb{F}_2^n)$

$$g : \mathbb{F}_2^{2n} \rightarrow \mathbb{F}_2^{2n+1}$$

Output: $(\mathbf{r}, f(\mathbf{s}), \mathbf{r} \cdot \mathbf{s})$

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Exercise:

Step 1: for $\frac{\gamma}{2}$ fraction of s , we have:

$$\Pr_{\mathbf{r}}[A(\mathbf{r}, f(s)) = \mathbf{r} \cdot s] \geq \frac{1}{2} + \frac{\gamma}{2}$$

Restrict to these s in the subsequent steps.

Step 2: For $h(\cdot) = A(\cdot, f(s))$ and $S = \{i : s_i = 1\}$, we have:

also switching from $\mathbb{F}_2$ to $\{-1, 1\}$

$$\hat{h}(S) \geq \gamma$$

Step 3: Therefore, s can be learned using **Goldreich–Levin algorithm**.

Proof sketch

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$g : \mathbb{F}_2^{2n} \rightarrow \mathbb{F}_2^{2n+1}$

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Suppose an efficient algorithm A can predict bit $\mathbf{r} \cdot \mathbf{s}$ from $\mathbf{r}, f(\mathbf{s})$ with non-negligible advantage:

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Step 3: Therefore, $\mathbf{s}$ can be learned using **Goldreich–Levin algorithm**.

g is a pseudorandom generator.

Such an algorithm A does not exist.

Contradicting the assumption that f is a one-way permutation.

References

- Sections 3.3 and 3.5 of <https://arxiv.org/abs/2105.10386>